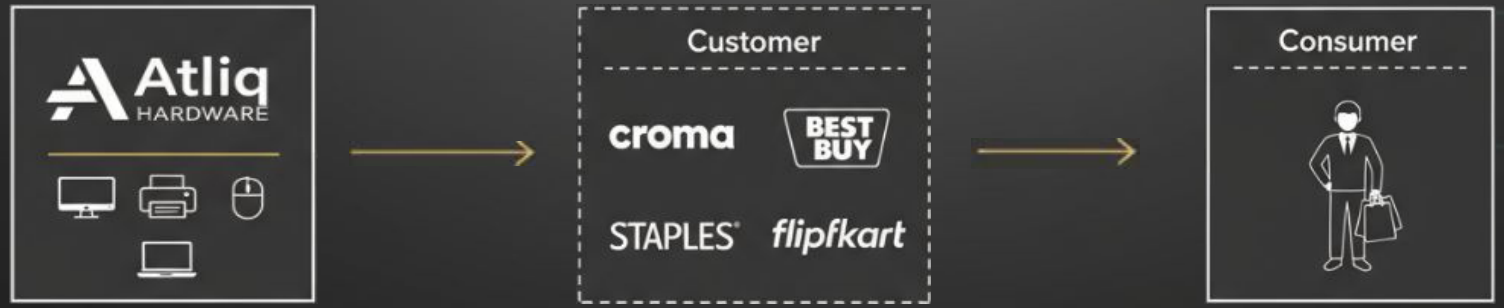
AtliQ Supply Chain & Finance Analytics Mega Project

Supply Chain Finance Management

Introduction

AtliQ Technology, established in 2018, operates in the hardware manufacturing sector, specializing in electronic gadgets such as mice, keyboards, laptops, and printers. The company distributes products to a global network of customers, including both brick-and-mortar and online stores.

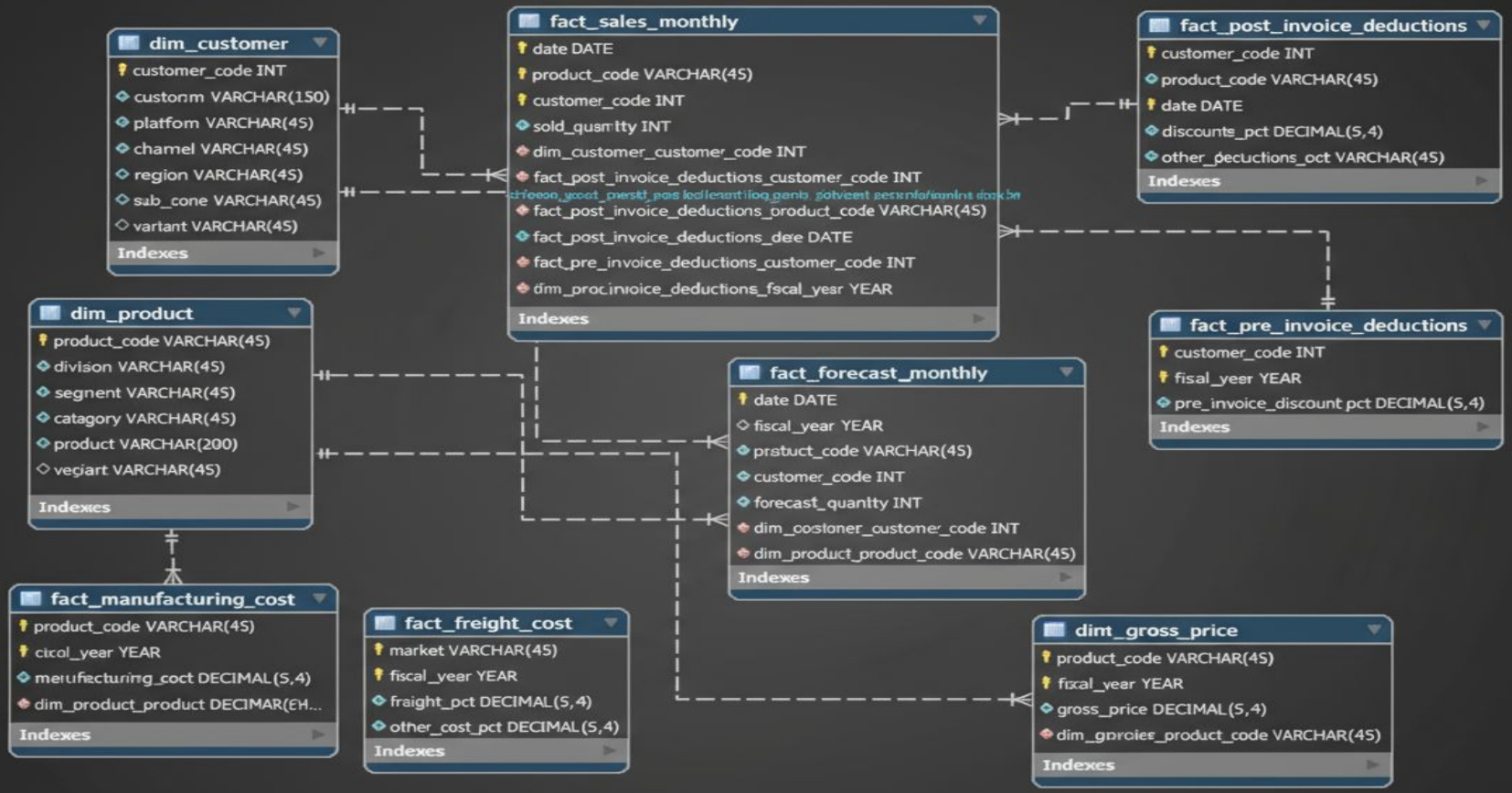
The objective of this project is to leverage SQL-based analytics to unlock actionable business insights for AtliQ, improve financial and supply chain performance, and drive strategic decision-making.



Methodologies

- Designed and built a star schema-based data warehouse using MS SQL Server, consisting of fact and dimension tables.
- Populated tables with transactional, customer, product, pricing, discounts, and forecasting data.
- Wrote advanced SQL queries using aggregations, window functions, user-defined functions, stored procedures, and triggers.
- Conducted analysis for sales trends, forecast accuracy, cost reporting, and inventory optimization.

Methodologies - An Entity Relationship Diagram (ERD)



Approaches

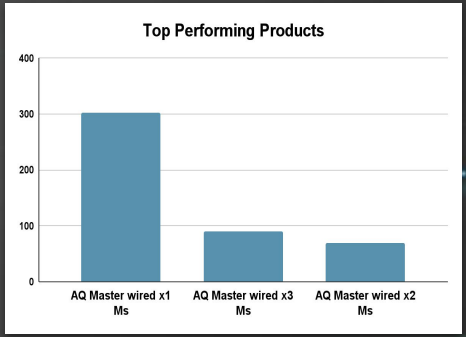
- Advanced use of window functions (ROW_NUMBER, RANK, LAG, LEAD, etc.) to analyze sales ranks, growth rates, and outliers.
- Used dynamic pivot queries for comparative monthly analysis.
- Developed triggers to automate business processes (inventory updates, audit logs, constraint enforcement).
- Created user-defined functions for reusable business logic (e.g., average discounts, total freight cost).

```sql
SELECT *
FROM
(
    SELECT
        f.fiscal_year,
        DATENAME(MONTH, f.date) AS Month,
        SUM(f.forecast_quantity) AS Total_Forecast,
        SUM(s.sold_quantity) AS Total_Sold,
        CASE
            WHEN SUM(f.forecast_quantity) > 0
            THEN ROUND(100.0 * SUM(s.sold_quantity) / SUM(f.forecast_quantity), 2)
            ELSE NULL
        END AS Forecast_Accuracy_Pct
    FROM
        fact_forecast_monthly f
        LEFT JOIN fact_sales_monthly s
            ON f.product_code = s.product_code
            AND f.customer_code = s.customer_code
            AND MONTH(f.date) = MONTH(s.date)
            AND YEAR(f.date) = YEAR(s.date)
    WHERE
        f.product_code = 'A0118150101'
    GROUP BY
        f.fiscal_year,
        DATENAME(MONTH, f.date)
) SourceTable
PIVOT
(
    MAX(Forecast_Accuracy_Pct)
    FOR Month IN ([January], [February], [March], [April], [May], [June], [July], [August], [September], [October], [November], [December])
) AS PivotedTable;
```

Key Findings

- Identified top-performing products and customers by total sales and revenue.
- Detected significant variances in forecast accuracy across months and products.
- Pinpointed periods of over-discounting impacting gross margin.
- Mapped freight and manufacturing costs by market, highlighting cost-saving opportunities.
- Found recurring seasonal sales patterns and customer loyalty trends.

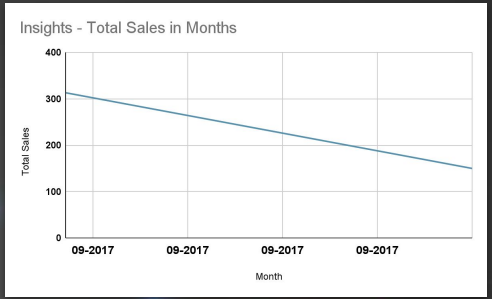
	Product	Total_Sold
1	AQ Master wired x1 Ms	303
2	AQ Master wired x3 Ms	90
3	AQ Master wired x2 Ms	70



Average of Accuracy_Percent	Column Labels	
Row Labels	Month	Grand Total
AQ Master wired x1 Ms	-175.3314876	-175.3314876
Grand Total	-175.3314876	-175.3314876

Insights

- Products with highest sales are not always the most profitable due to discounts and varying costs.
- Forecast accuracy is highest in established markets, lower for new launches.
- High freight costs in certain markets suggest a need for logistics optimization.
- Seasonal spikes (Sept–Dec) drive sales boosts; inventory must align with forecast.
- Direct sales channels yield higher gross margins than traditional distribution in this segment.



Actionable Recommendations

- Adjust forecast models based on historical error patterns and customer segmentation.
- Reconsider discount strategy for underperforming products to maximize margin.
- Negotiate for better freight contracts or assess alternate logistics partners in high-cost markets.
- Enhance inventory planning to reduce overstock during low-sales months.
- Increase focus on direct sales channels in high-margin regions.

Focus Area	Current State (from data)	Recommended State/Action
Forecast Accuracy	Varies by product; some months/products <80%	Use historical error to adjust models for low-performing items
Discount Strategy	Several SKUs with excessive discounts (often >20%)	Limit discounts to <15% unless proven margin gain
Freight/Logistics	High cost in some markets (e.g., EU, South Asia)	Renegotiate freight contracts; prioritize optimization in high-cost zones
Inventory Planning	Overstock seen in Marâ€“Jun; understock in Novâ€“Dec	Sync inventory with seasonal sales, adjust safety stock by market
Channel Margins	Direct e-commerce channels more profitable	Increase share of direct sales in high margin regions

Conclusions

- The integration of advanced SQL analytics empowers AtliQ to make data-driven supply chain and finance decisions.
- Key business questions about profitability, forecasting, and customer value are now answerable in real-time.
- Continued investment in data infrastructure and analytics capability is crucial for sustained competitive advantage.